

.NET Lab: CLR Startup & Heap Monitoring

Duration : 30 Min.

Scenario

You need to baseline the memory usage and startup execution flow of a new .NET Core Web API service.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Launch the service, use `dotnet-counters` to monitor Gen 0, 1, and 2 allocation rates under baseline traffic, and save the metrics log report.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. CLR process launch and monitoring commands config: 5 marks
2. Gen 0, 1, 2 allocation metrics output reports validation: 5 marks

Total Score: 10 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.